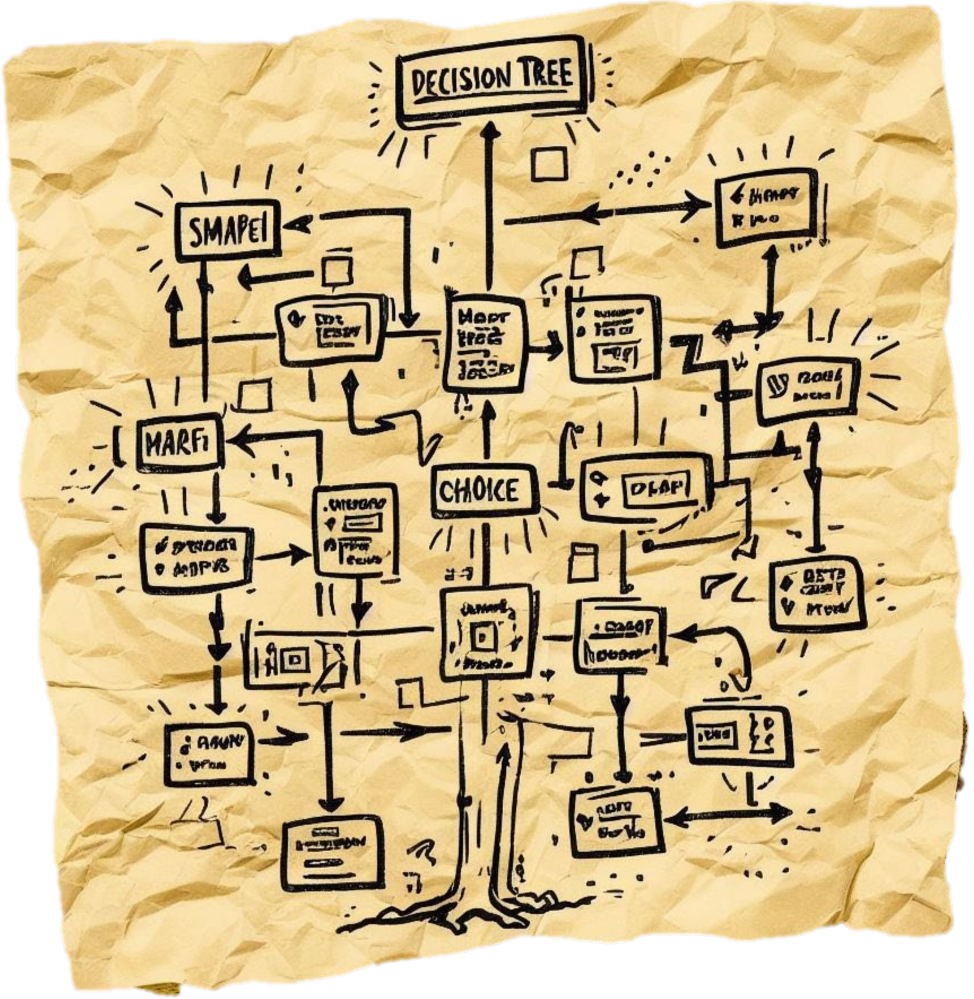
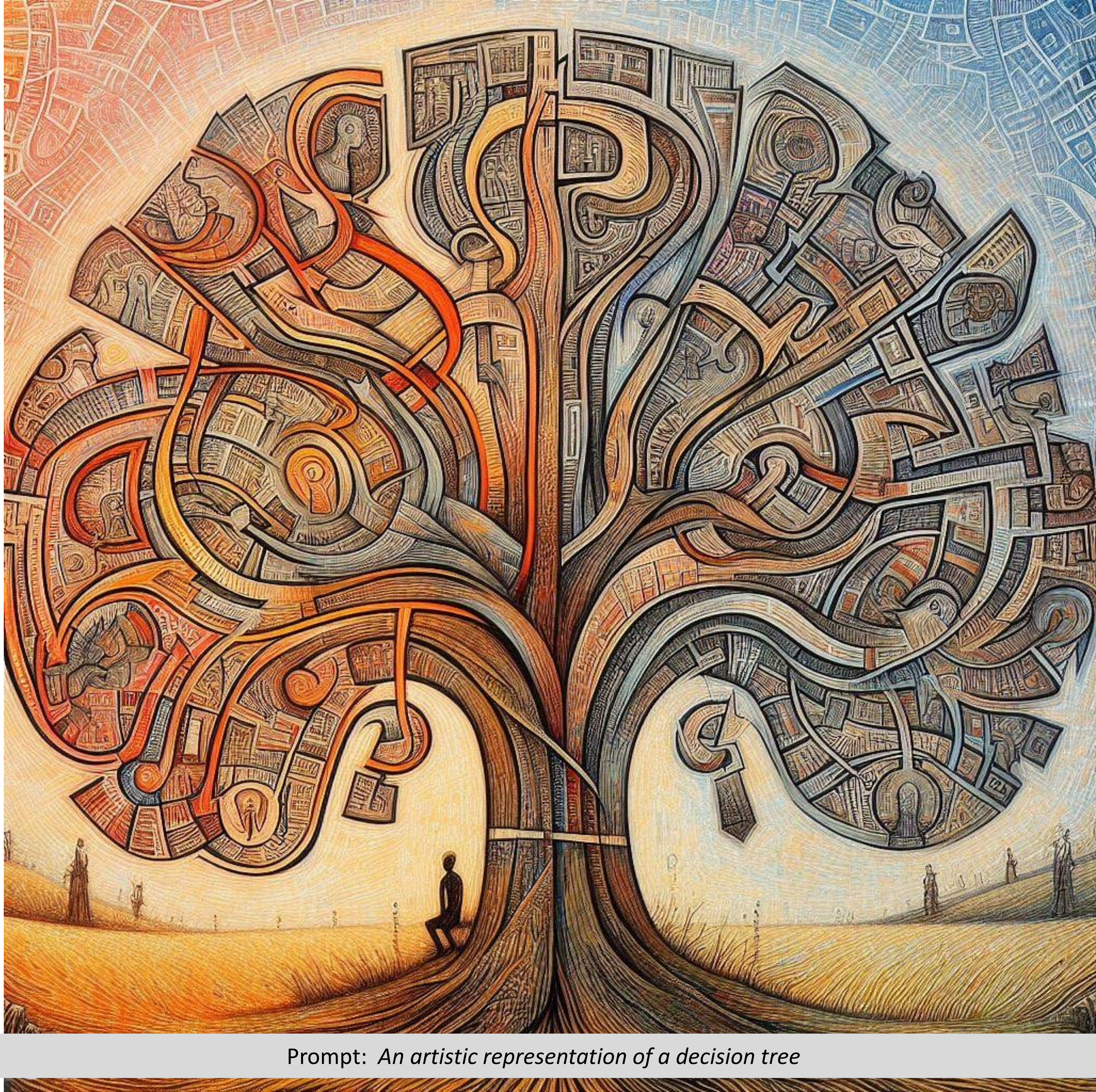


Decision Tree and Random Forest

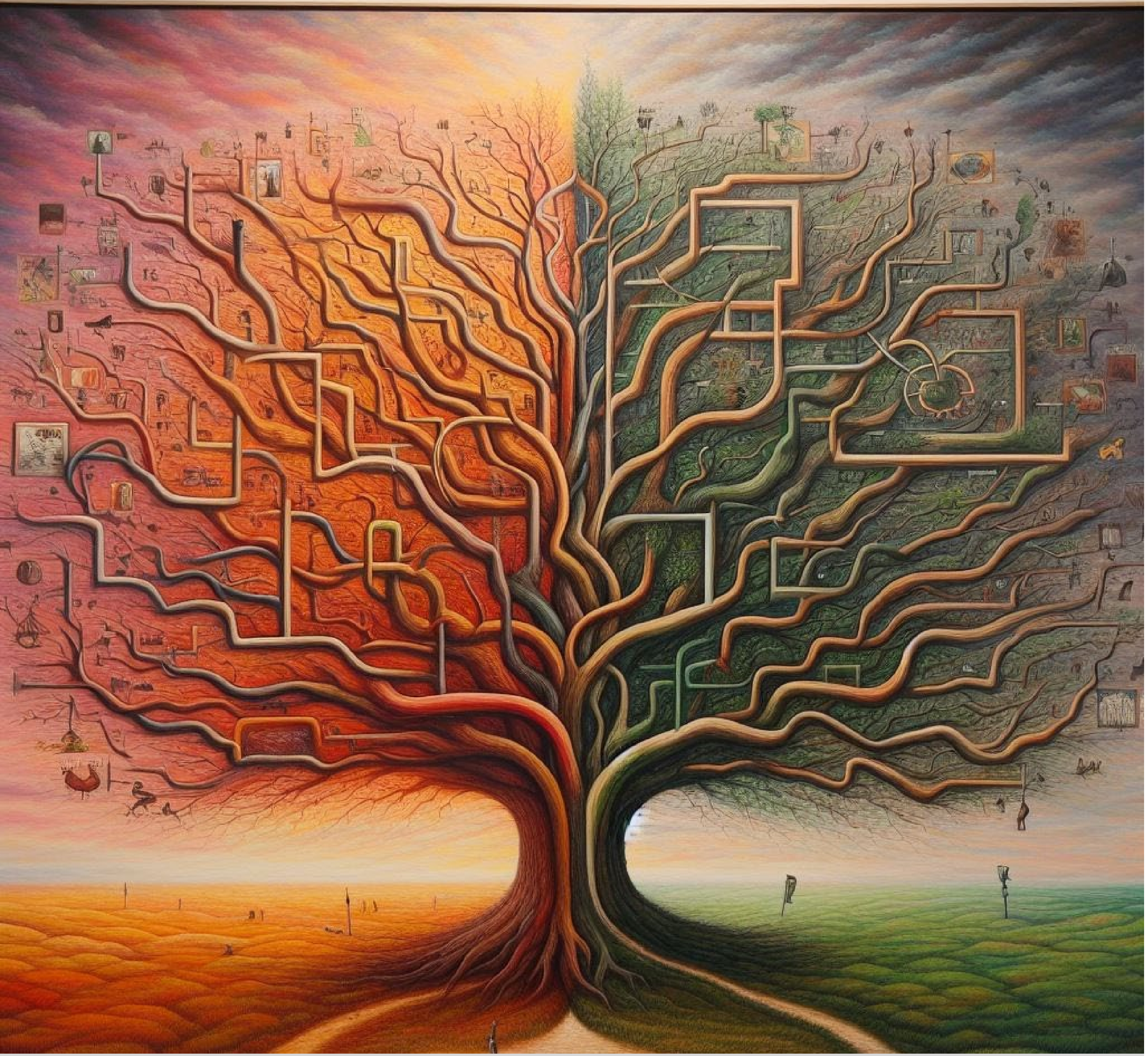
Francesco Carrino

Outline

- Introduction
  - Lazy learning
  - Eager learning
- Decision Tree
- Random Forest



Prompt: *An artistic representation of a decision tree*



Prompt: *An artistic representation of a decision tree*

Decision Tree

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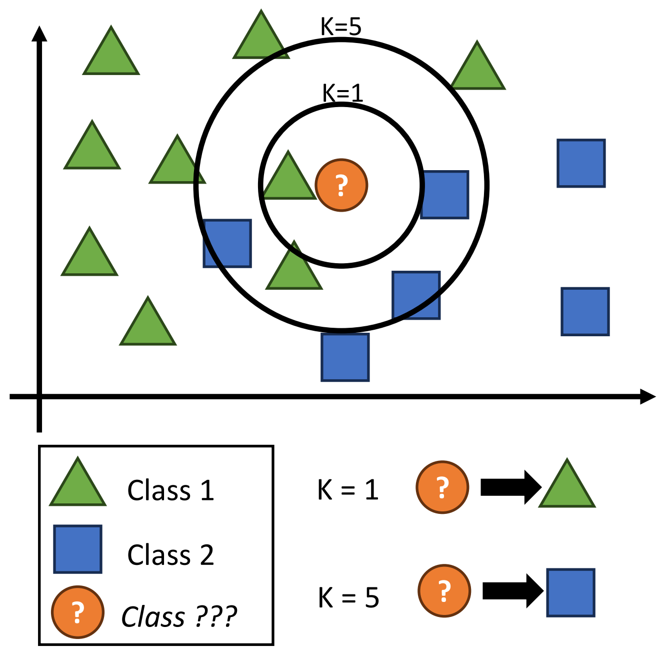
Lazy learning Vs Eager learning

- Lazy learning
  - Does not create a generalized model during the training phase
  - The model postpones generalization until a new instance needs to be classified or predicted
  - Needs the original training data to make predictions
- Example: k-Nearest Neighbors

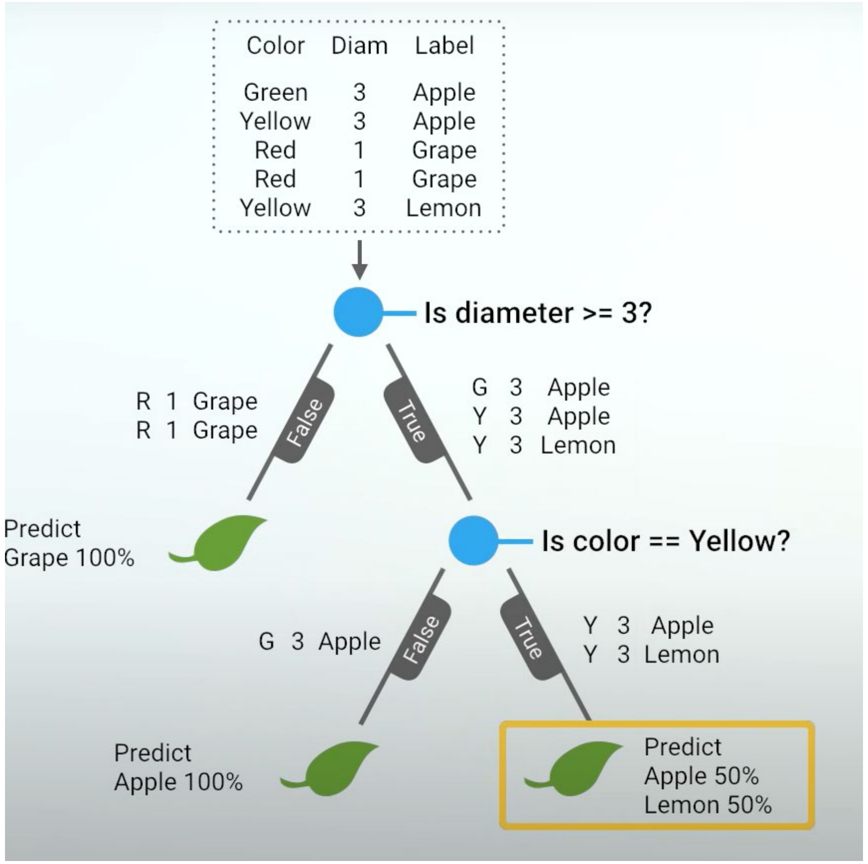
- Eager learning
  - Builds a generalized, static representation of the training data during the training phase
  - Creates a summary or structure of the data, and this structure is used for prediction
  - (Once trained) Does not need the original training data to make predictions
- Example: **Decision trees**, support vector machines, and neural networks

Lazy learning Vs Eager learning

- Lazy learning



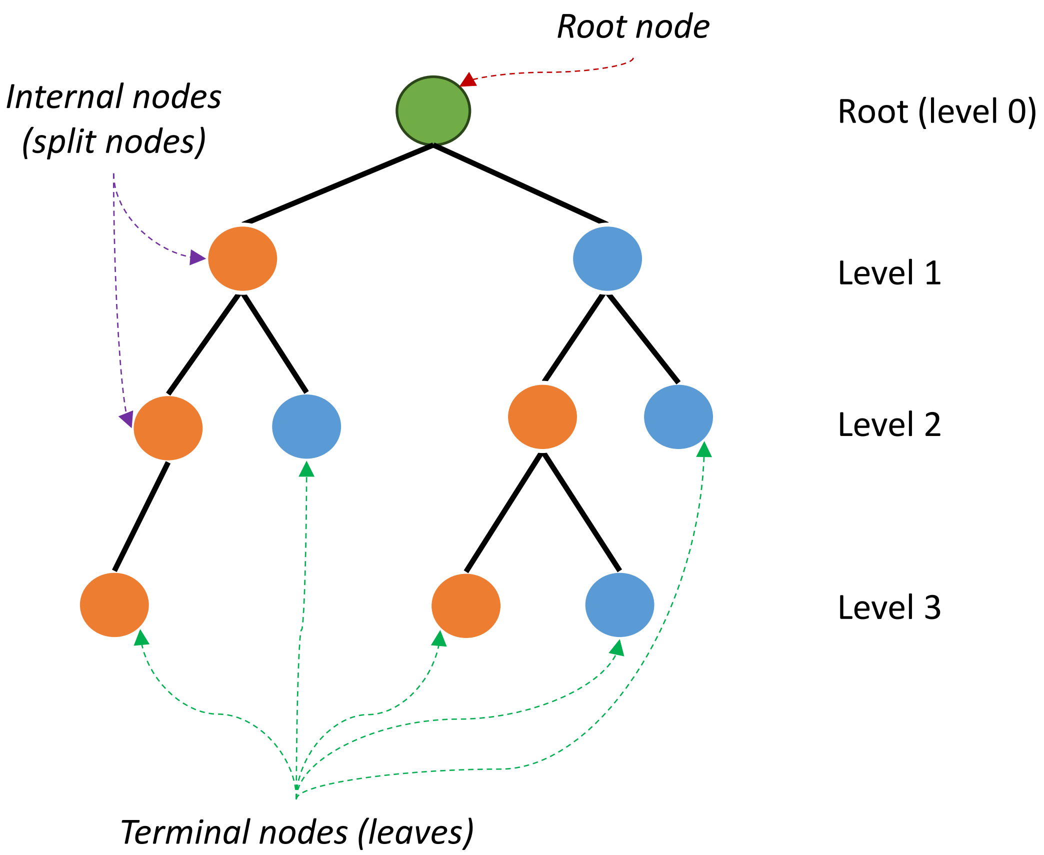
- Eager learning



Source: Let's Write a Decision Tree Classifier from Scratch
(Google for Developers)

Trees

and binary trees



- A **tree** is a set of nodes and edges organized hierarchically
  - Root node
  - Internal node (or “split node”)
  - Terminal node (or “leaf”)
  - Edge (connection between nodes)

- Nodes and edges - Each node has only 1 incoming edge but can have multiple outgoing edges
  - Binary tree: 2 outgoing edges
  - Ternary tree: 3 outgoing edges
  - Heterogeneous tree : 2,3,… outgoing edges

- **Note**: any tree can be transformed into a binary tree (binary tree encoding)

Decision Tree

- Decision Tree is a *supervised* machine learning algorithm used for both *classification* and *regression* tasks.

- **Idea**: It recursively splits the dataset into subsets based on the most significant attribute(s) at each node, creating a tree-like model of decisions.

Decision tree

A definition

Decision Tree

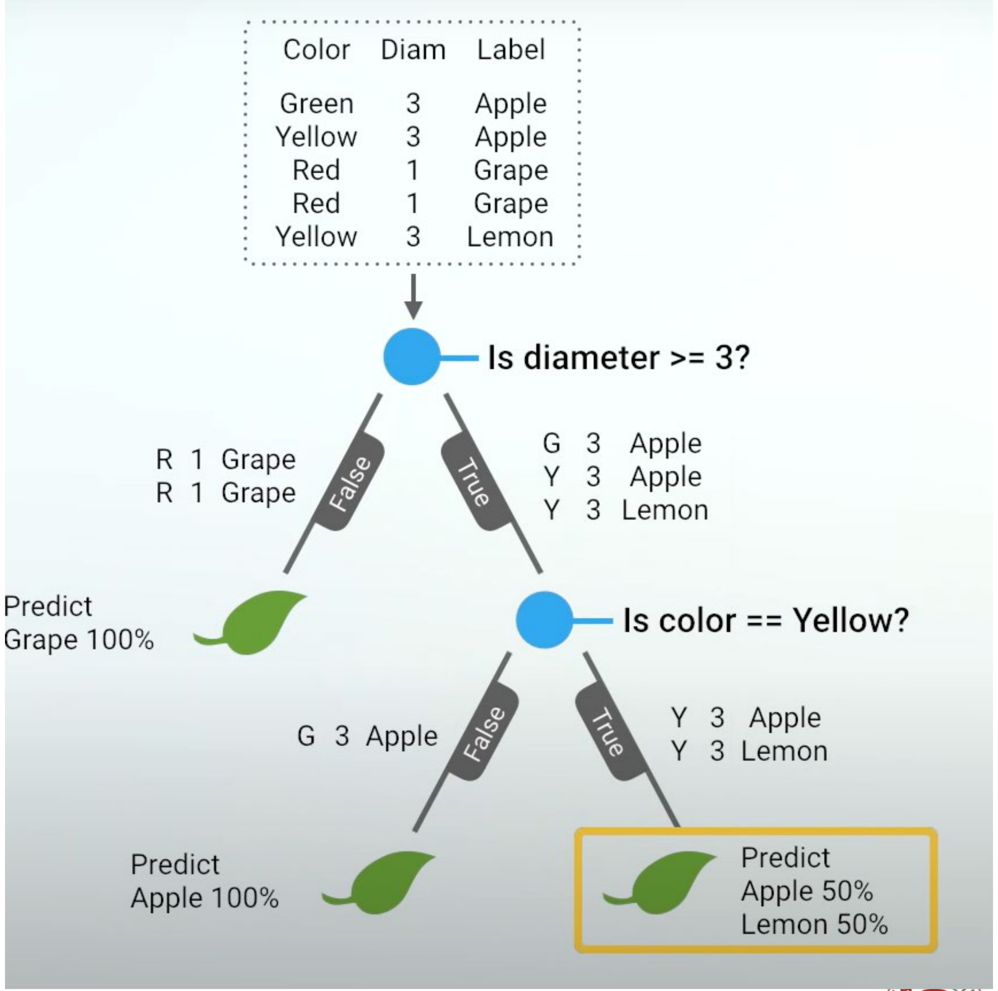
“A decision tree is a ***flowchart-like*** *structure in which each* ***internal node represents a "test"*** *on an attribute, each* ***branch represents the outcome of the test****, and each* ***leaf node represents a class label*** *(or prediction). The* ***paths from root to leaf represent the rules****.”*

~Wikipedia

Decision tree

For classification

- The **root** represents the entire dataset
- **Internal nodes** represent a “test” on an attribute
- **Each branch** represents the outcome of the test
- **Each leaf** node represents a class label (decision taken after computing all attributes).
- **The paths** from root to leaf represent classification rules



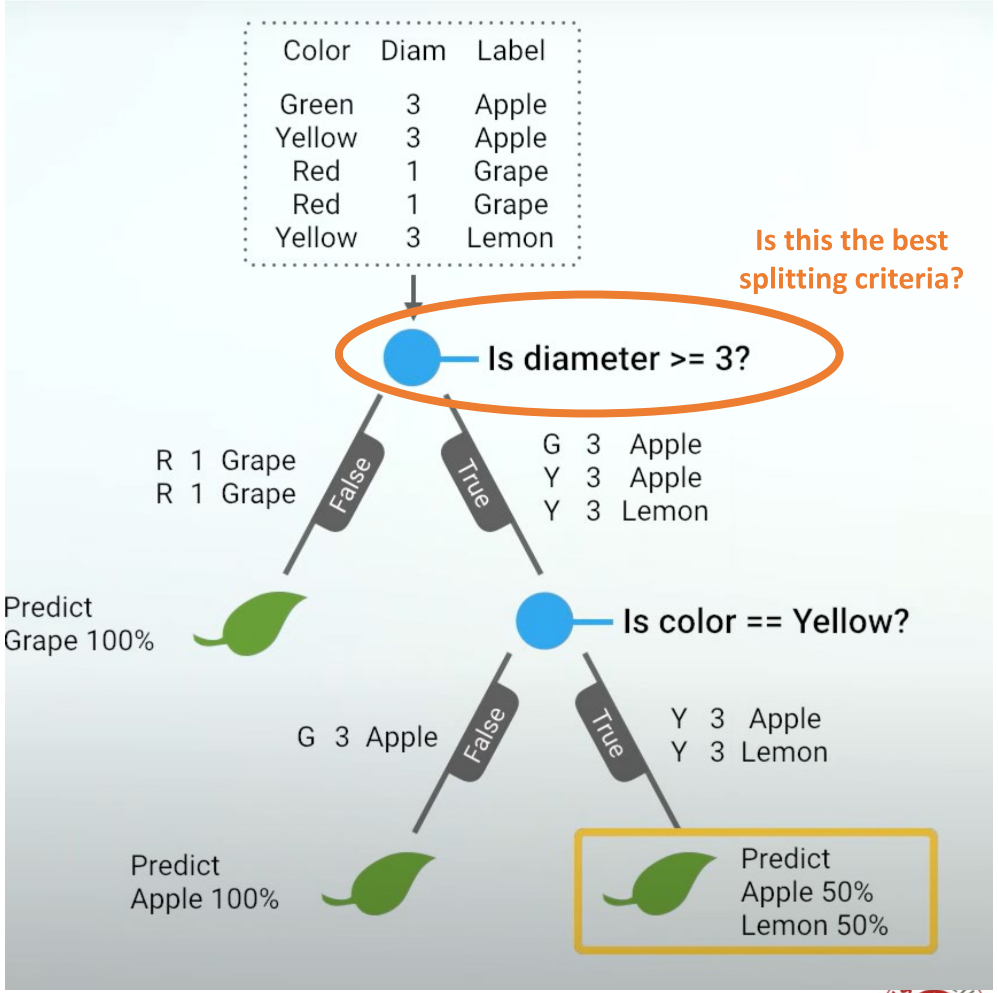
Source: [Let’s Write a Decision Tree Classifier from Scratch](#) (Google for Developers)

Decision tree

How it works

- Decision tree is an Eager learning algorithm:
  - It builds a generalized, static representation of the training data during the training phase
- During the training phase, the algorithm selects the ***optimum*** splitting criteria (*for a given training set*) that provides the **best separation** of the data

- How?



Source: Let's Write a Decision Tree Classifier from Scratch (Google for Developers)

Hes·so // VALAIS WALLIS
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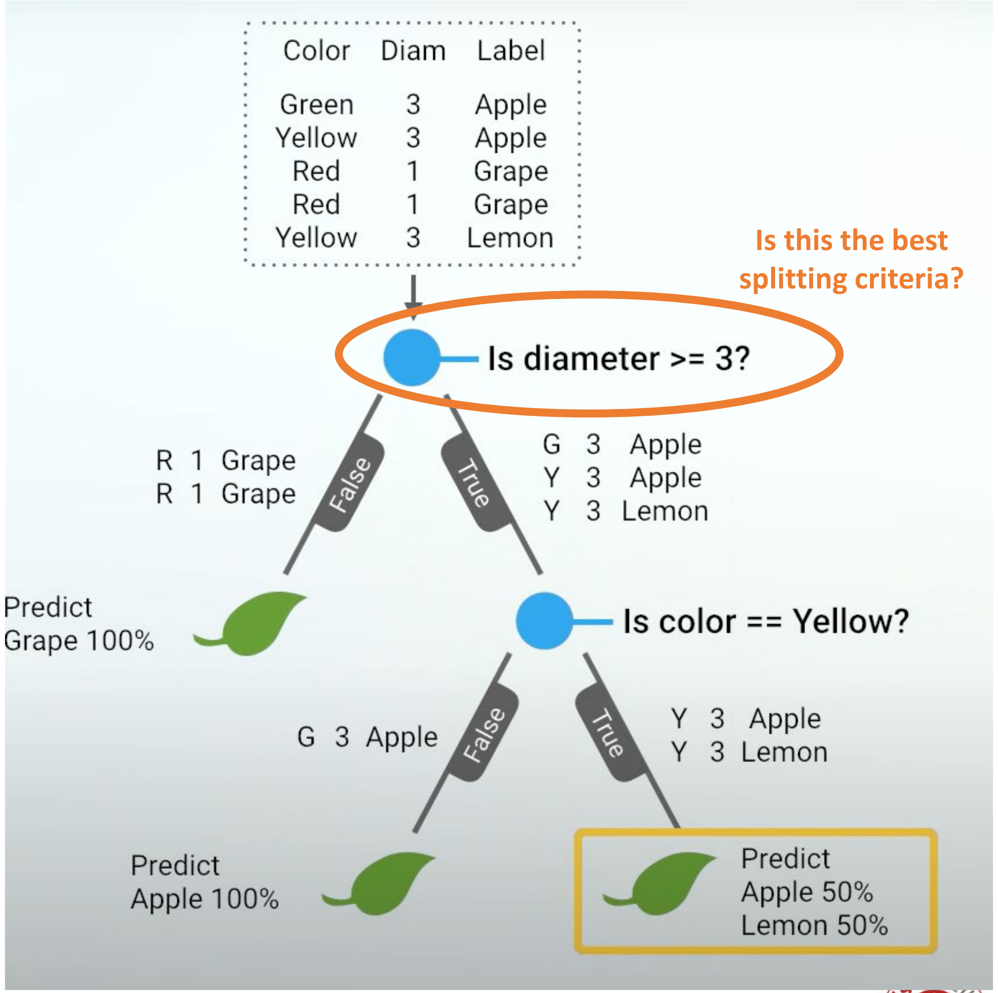
Decision tree

How it works

- The questions are:
  - What is a good separation?
  - How can we quantify it?
  - How can we maximize it?

- The answer: “uncertainty”
  - Gini impurity
  - (or entropy, or log loss…)



Source: [Let’s Write a Decision Tree Classifier from Scratch](#) (Google for Developers)

Creating the tree

Gini impurity

- Gini impurity evaluates how often a **randomly chosen** element would be **incorrectly** labeled if it was randomly labeled according to the distribution of labels in a set.

$$Gini\ impurity(node) = 1 - \sum_{i=1}^{c} p_i^2$$

- Where:
  - c is the number of classes (categories or labels)
  - p_i^2 is the probability of randomly picking an element of class i in the node

Creating the tree

Gini impurity - Interpretation

$$Gini\ impurity(node) = 1 - \sum_{i=1}^{c} p_i^2$$

- **Where**:
  - c is the number of classes (categories or labels)
  - p_i^2 is the probability of randomly picking an element of class i in the node

- **Interpretation**: Gini impurity ranges from 0 to 1
  - 0 indicates that the node is pure (all elements belong to the same class)
  - 1 indicates maximum impurity (elements are evenly distributed across all classes)

Creating the tree

Example: I Root node

- Let's compute the Gini impurity for the root node (before any split).
  1. Calculate class probability

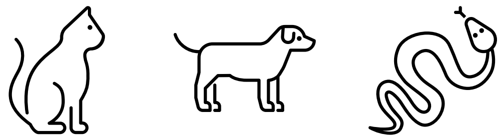
$$p_{cat} = \frac{4}{10} \quad p_{dog} = \frac{3}{10} \quad p_{snake} = \frac{3}{10}$$

  2. Compute Gini impurity

$$Gini\ impurity(root) = 1 - \left(\left(\frac{4}{10}\right)^2 + \left(\frac{3}{10}\right)^2 + \left(\frac{3}{10}\right)^2\right)$$

$$Gini\ impurity(root) = 1 - \left(\frac{16}{100} + \frac{9}{100} + \frac{9}{100}\right)$$

$$Gini\ impurity(root) = 1 - \frac{34}{100} = \frac{66}{100} = 0.66$$



Label	Weight	Height	Number of legs	Class (biology)
Cat	5	10	4	Mammalia
Dog	10	15	4	Mammalia
Snake	2	3	0	Reptilia
Cat	6	12	4	Mammalia
Dog	12	18	4	Mammalia
Snake	1	2	0	Reptilia
Cat	4	8	4	Mammalia
Dog	8	14	4	Mammalia
Snake	3	5	0	Reptilia
Cat	7	14	4	Mammalia

Reminder: $Gini\ impurity\ (node) = 1 - \sum_{i=1}^{c} p_i^2$

Creating the tree

Information gain

General formulation

$$Information\ gain(split) = uncertainity(parent) - avg\_uncertainity_{weighted}(after\ the\ split)$$

Gini impurity used as uncertainty measure

$$Information\ gain(split) = Gini\ impurity(parent) - \sum\left(\underbrace{\frac{|S_i|}{|S|}}_{\text{Weight (relevance of a subset)}} * \underbrace{Gini\ impurity(S_i)}_{\text{Impurity of a subset}}\right)$$

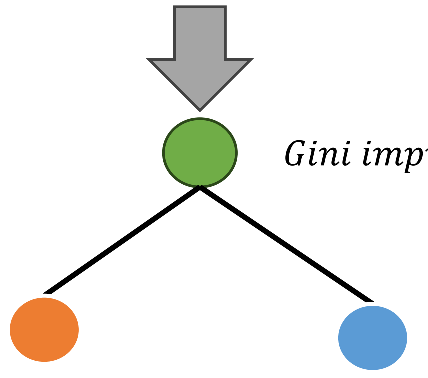
- **Where**:
  - $\sum$ denotes the sum over all subsets created by the split
  - $|S_i|$ is the number of instances in the subset S_i
  - $|S|$ is the total number of instances in the parent set

- **Interpretation**:
  - Information gained (uncertainty reduction) after a given split
  - The higher the information gain, the more effective the split is in terms of reducing uncertainty

Creating the tree

Example: II Split

Label	Weight	Height	Number of legs	Class (biology)
Cat	5	10	4	Mammalia
Dog	10	15	4	Mammalia
Snake	2	3	0	Reptilia
Cat	6	12	4	Mammalia
Dog	12	18	4	Mammalia
Snake	1	2	0	Reptilia
Cat	4	8	4	Mammalia
Dog	8	14	4	Mammalia
Snake	3	5	0	Reptilia
Cat	7	14	4	Mammalia



$$Gini\ impurity(root) = \mathbf{0.66}$$

Suppose: we split on the question:
Class (biology) == "Mammalia"

- How do we split?
  - We ask a question: True | False
- Which kind of "***question***" can we ask?
  - At each node: we iterate over *every value* for *every feature* (that appears on this node)
    - Is weight >= 5?
    - Is heigth >= 10
    - Is number_of_legs >= 4?
    - Is class == "Mammalia" ?
    - …
- How to choose the best question?
  - **Information gain!**

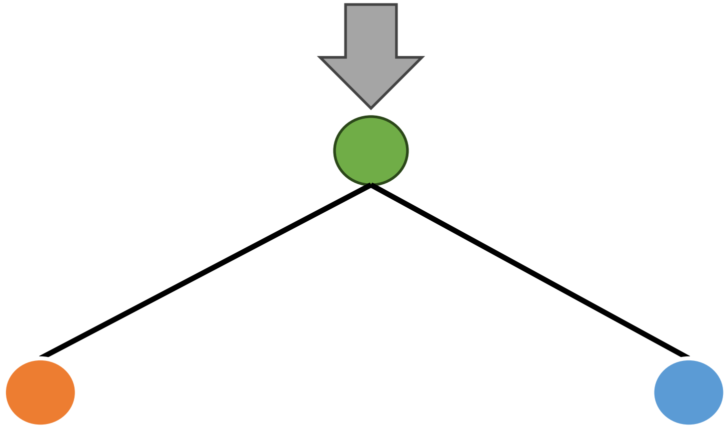
Creating the tree

Example: II Split

Label	Weight	Height	Number of legs	Class (biology)
Cat	5	10	4	Mammalia
Dog	10	15	4	Mammalia
Snake	2	3	0	Reptilia
Cat	6	12	4	Mammalia
Dog	12	18	4	Mammalia
Snake	1	2	0	Reptilia
Cat	4	8	4	Mammalia
Dog	8	14	4	Mammalia
Snake	3	5	0	Reptilia
Cat	7	14	4	Mammalia

$Gini\ impurity(root) = 0.66$

$|S| = 10$ (total number of instances in the parent set)



Suppose: we split on the question:
Class (biology) == "Mammalia"

Label	Weight	Height	Number of legs	Class (biology)
Cat	5	10	4	Mammalia
Dog	10	15	4	Mammalia
Cat	6	12	4	Mammalia
Dog	12	18	4	Mammalia
Cat	4	8	4	Mammalia
Dog	8	14	4	Mammalia
Cat	7	14	4	Mammalia

$$Gini\ impurity(orange\ node) = 1 - \left(\left(\frac{4}{7}\right)^2 + \left(\frac{3}{7}\right)^2 + \left(\frac{0}{7}\right)^2\right)$$

$$Gini\ impurity(orange\ node) = 1 - \left(\frac{25}{49}\right) = \mathbf{0.49}$$

$|S\_i| = 7$ (number of instances in this node)

Label	Weight	Height	Number of legs	Class (biology)
Snake	2	3	0	Reptilia
Snake	1	2	0	Reptilia
Snake	3	5	0	Reptilia

$$Gini\ impurity(blue\ node) = 1 - \left(\left(\frac{0}{3}\right)^2 + \left(\frac{0}{3}\right)^2 + \left(\frac{3}{3}\right)^2\right)$$

$$Gini\ impurity(blue\ node) = \mathbf{0}$$

$|S\_i| = 3$ (number of instances in this node)

- IDEA -
Nodes with more items are more important!

Creating the tree

Example: II Split

$Gini\ impurity(root) = \mathbf{0.66}$

$|S| = 10$ (total number of instances in the parent set)

Suppose: we split on the question:
Class (biology) == "Mammalia"

Label	Weight	Height	Number of legs	Class (biology)
Cat	5	10	4	Mammalia
Dog	10	15	4	Mammalia
Cat	6	12	4	Mammalia
Dog	12	18	4	Mammalia
Cat	4	8	4	Mammalia
Dog	8	14	4	Mammalia
Cat	7	14	4	Mammalia

$Gini\ impurity(orange\ node) = \mathbf{0.49}$

$|S_{orange}| = 7$ (number of instances in this node)

- IDEA -
Nodes with more items are more important!

Label	Weight	Height	Number of legs	Class (biology)
Snake	2	3	0	Reptilia
Snake	1	2	0	Reptilia
Snake	3	5	0	Reptilia

$Gini\ impurity(blue\ node) = \mathbf{0}$

$|S_{blue}| = 3$ (number of instances in this node)

$$Information\ gain(split) = Gini\ impurity(parent) - \sum\left(\frac{|S_i|}{|S|} * Gini\ impurity(S_i)\right)$$

$$Information\ gain(split) = 0.66 - \left(\frac{7}{10} * 0.49 + \frac{3}{10} * 0\right)$$

$$Information\ gain(split) = \mathbf{0.3234}$$

Creating the tree

Maximizing the $Information\ gain$

- At each node: we iterate over *every value* for *every feature* (that appears on this node)
- We choose the split that (for that node) maximize the information gain
- And repeat!

Exercise

Maximizing the *Information gain*

- Compute the information gain for the split: weight >= 6?
- Would it be a better or worse *first* split?

Label	Weight	Height	Number of legs	Class (biology)
Cat	5	10	4	Mammalia
Dog	10	15	4	Mammalia
Snake	2	3	0	Reptilia
Cat	6	12	4	Mammalia
Dog	12	18	4	Mammalia
Snake	1	2	0	Reptilia
Cat	4	8	4	Mammalia
Dog	8	14	4	Mammalia
Snake	3	5	0	Reptilia
Cat	7	14	4	Mammalia

Completing the tree

Label	Weight	Height	Number of legs	Class (biology)
Cat	5	10	4	Mammalia
Dog	10	15	4	Mammalia
Snake	2	3	0	Reptilia
Cat	6	12	4	Mammalia
Dog	12	18	4	Mammalia
Snake	1	2	0	Reptilia
Cat	4	8	4	Mammalia
Dog	8	14	4	Mammalia
Snake	3	5	0	Reptilia
Cat	7	14	4	Mammalia

Class (biology) == "Mammalia"

Y

Label	Weight	Height	Number of legs	Class (biology)
Cat	5	10	4	Mammalia
Dog	10	15	4	Mammalia
Cat	6	12	4	Mammalia
Dog	12	18	4	Mammalia
Cat	4	8	4	Mammalia
Dog	8	14	4	Mammalia
Cat	7	14	4	Mammalia

N

Label	Weight	Height	Number of legs	Class (biology)
Snake	2	3	0	Reptilia
Snake	1	2	0	Reptilia
Snake	3	5	0	Reptilia

weight >= 8

Y

Label	Weight	Height	Number of legs	Class (biology)
Dog	10	15	4	Mammalia
Dog	12	18	4	Mammalia
Dog	8	14	4	Mammalia

N

Label	Weight	Height	Number of legs	Class (biology)
Cat	5	10	4	Mammalia
Cat	6	12	4	Mammalia
Cat	4	8	4	Mammalia
Cat	7	14	4	Mammalia

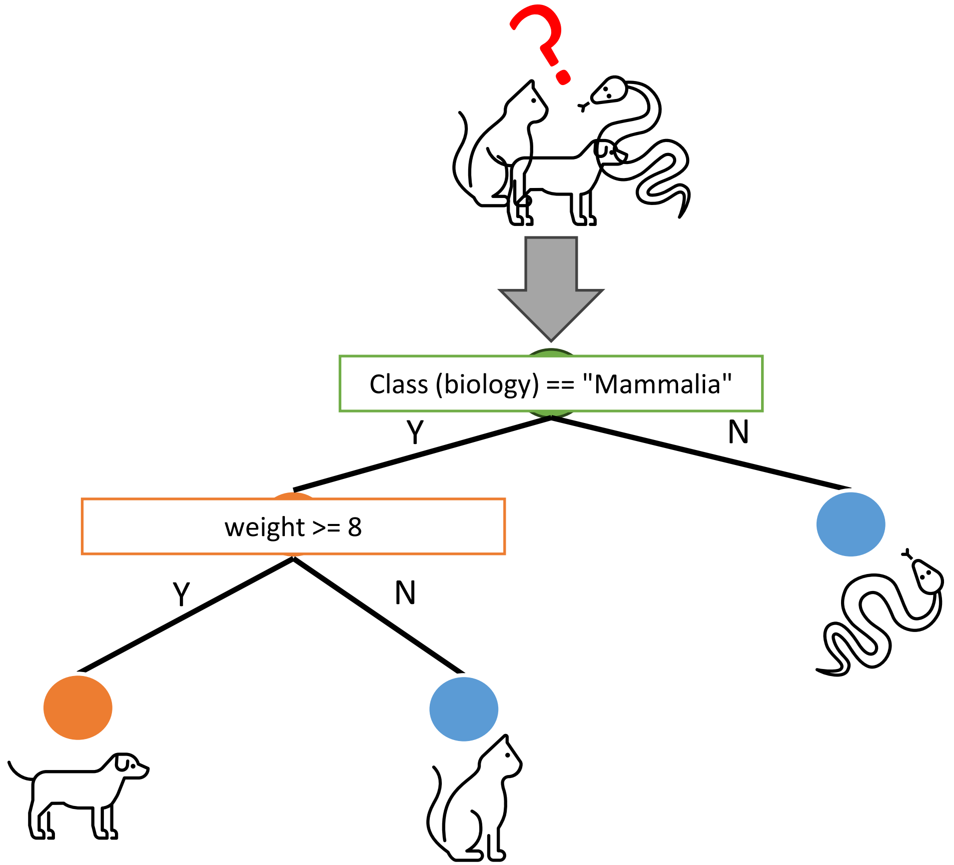
- Result -
With two split
we perfectly
separated the
dataset

Decision Tree

Result

- The so built Decision tree defines a simple flow chart that we can follow to classify new unseen sample

Label	Weight	Height	Number of legs	Class (biology)
??	5	10	4	Mammalia
??	7	12	4	Mammalia



Decision tree

Advantages

- **Interpretability**: Decision trees are easily interpretable and can be visualized graphically. The decision-making process is clear and understandable.
- **No Need for Feature Scaling**: Unlike some other algorithms, decision trees are not sensitive to the scale of features. There’s no need to standardize or normalize input features.
- **Handling Both Numerical and Categorical Data**: Decision trees can handle both numerical and categorical data without the need for one-hot encoding.
- **Automated Feature Selection**: The algorithm automatically selects the most relevant features for splitting, reducing the need for feature engineering.
- **Nonlinear Relationships**: Decision trees can capture nonlinear relationships between features and the target variable.
- **Low Prediction Cost**: Once trained, making predictions using a decision tree is computationally inexpensive.

Decision tree

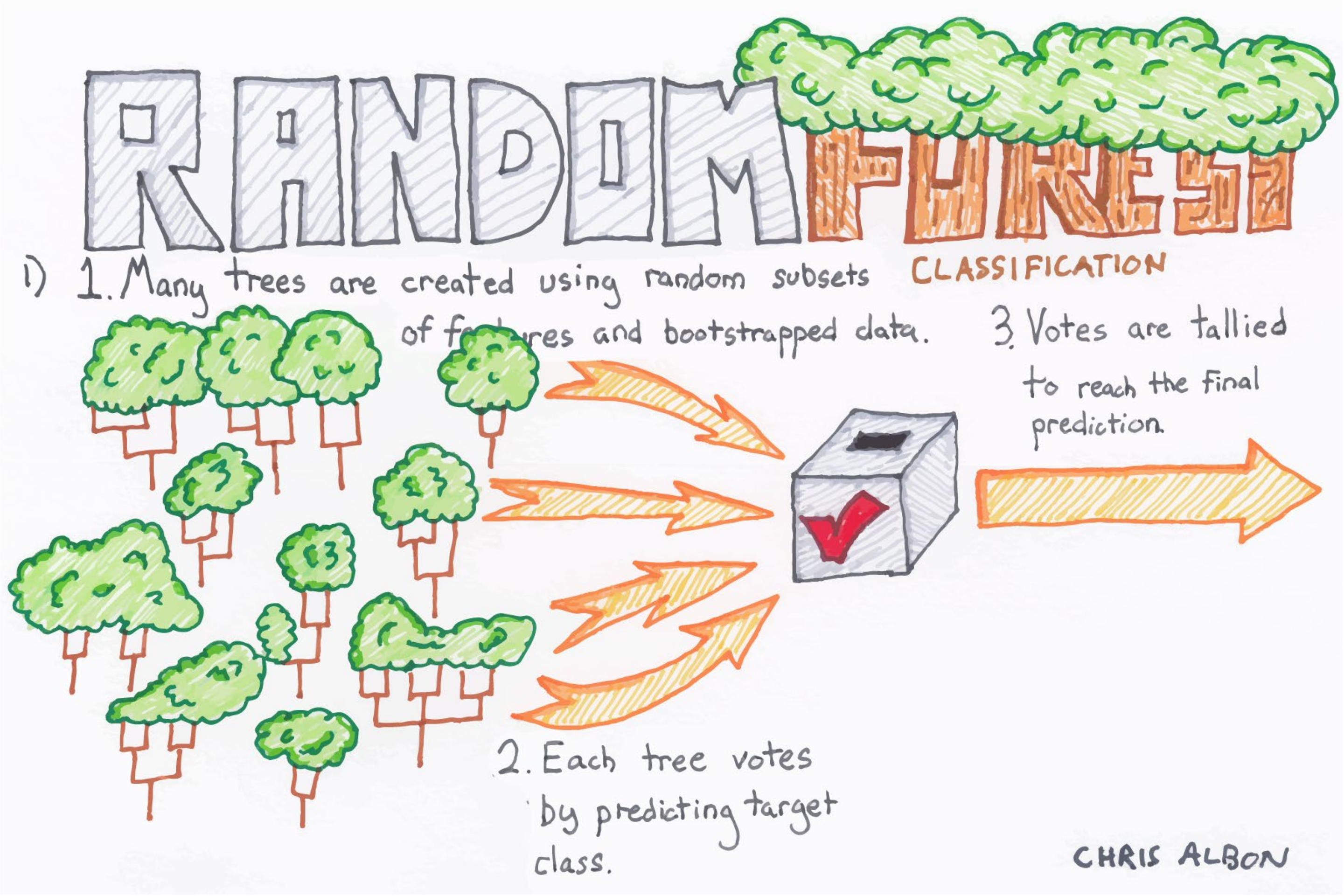
Drawbacks

- **Overfitting**: Decision trees are prone to overfitting, especially when the tree is deep and captures noise in the training data.
- **Instability**: Small changes in the data can lead to significantly different tree structures. This makes decision trees less stable compared to some other algorithms.
- **Biased to Dominant Classes**: In classification problems with imbalanced classes, decision trees can be biased towards the dominant class.
- **Greedy Nature**: Decision trees make locally optimal decisions at each node, which may not lead to a globally optimal tree. A globally optimal tree would require evaluating all possible splits, which is computationally expensive.
- **Limited Expressiveness**: A single decision tree may not capture complex relationships in the data as effectively as more sophisticated algorithms.
- **Exponential Growth**: The number of nodes and branches can grow exponentially with the depth of the tree, leading to complex models that may not generalize well.
- **Not Good for XOR-Like Relationships**: Decision trees may struggle to learn XOR-like relationships in the data.

Decision tree

Going deeper

- Pruning:
  - reducing the size of a Decision tree by removing sections of the tree that are non-critical and redundant to classify instances
  - https://en.wikipedia.org/wiki/Decision_tree_pruning
- Decision tree for regression:
  - Instead of predicting classes, they predict a continuous value.
  - Instead of maximizing the *information gain*, the objective during the splitting process is to reduce the variance of the target variable within each node, aiming to create partitions that are more homogenous with respect to the target variable.
  - https://scikit-learn.org/stable/modules/tree.html#regression
  - https://www.youtube.com/watch?v=g9c66TUylZ4



Source: https://gaussian37.github.io/ml-concept-RandomForest/

Random Forest

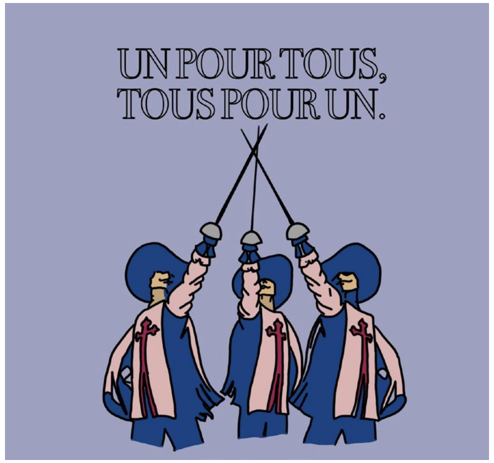
Introduction

Ensemble techniques

Ensemble learning

*“Ensemble methods use **multiple learning algorithms** to obtain **better predictive** performance than could be obtained from any of the constituent learning algorithms alone.”*

Wikipedia



- Switzerland’s motto: “*Unus pro Omnibus, Omnes pro Uno*”

Random Forest

Definition

- Random Forest
  - Ensemble learning method used for both classification and regression tasks
  - It builds multiple decision trees and merges their outputs to improve accuracy and reduce overfitting.

Random Forest

How it works

1. **Bootstrapped Sampling**: Random Forest starts by creating multiple “bootstrap samples” (random samples with replacement) from the original dataset.
   Each sample is used to train a separate decision tree.
2. **Random Feature Selection**: For each tree in the forest, a random subset of features is considered at each split.
   This helps to ensure diversity among the trees and prevents a single dominant feature from overly influencing the model.
3. **Decision Tree Construction**: A decision tree is constructed for each bootstrap sample.
   The tree is grown until a stopping criterion is met, such as a maximum depth or a minimum number of samples in a leaf.
4. **Voting (for Classification) or Averaging (for Regression)**:
   For classification tasks, the outputs of individual trees (class predictions) are combined through a majority vote.
   For regression tasks, the outputs are averaged.

Random Forest

How it works

Random Forest

How it works - example

1. **Bootstrapped Sampling**: Random Forest starts by creating multiple "bootstrap samples" (random samples with replacement) from the original dataset. Each sample is used to train a separate decision tree.
   - N samples = number of decision tree
   - All the samples usually have the same size but no necessarily the size of the full training set
   - Not necessarily contiguous

Label	Weight	Height	Number of legs	Class (biology)
Cat	5	10	4	Mammalia
Dog	10	15	4	Mammalia
Snake	2	3	0	Reptilia
Cat	6	12	4	Mammalia
Dog	12	18	4	Mammalia
Snake	1	2	0	Reptilia
Cat	4	8	4	Mammalia
Dog	8	14	4	Mammalia
Snake	3	5	0	Reptilia
Cat	7	14	4	Mammalia

Sample 1

Sample 3

.
.
.

Sample N

Sample 2

Random Forest

How it works - example

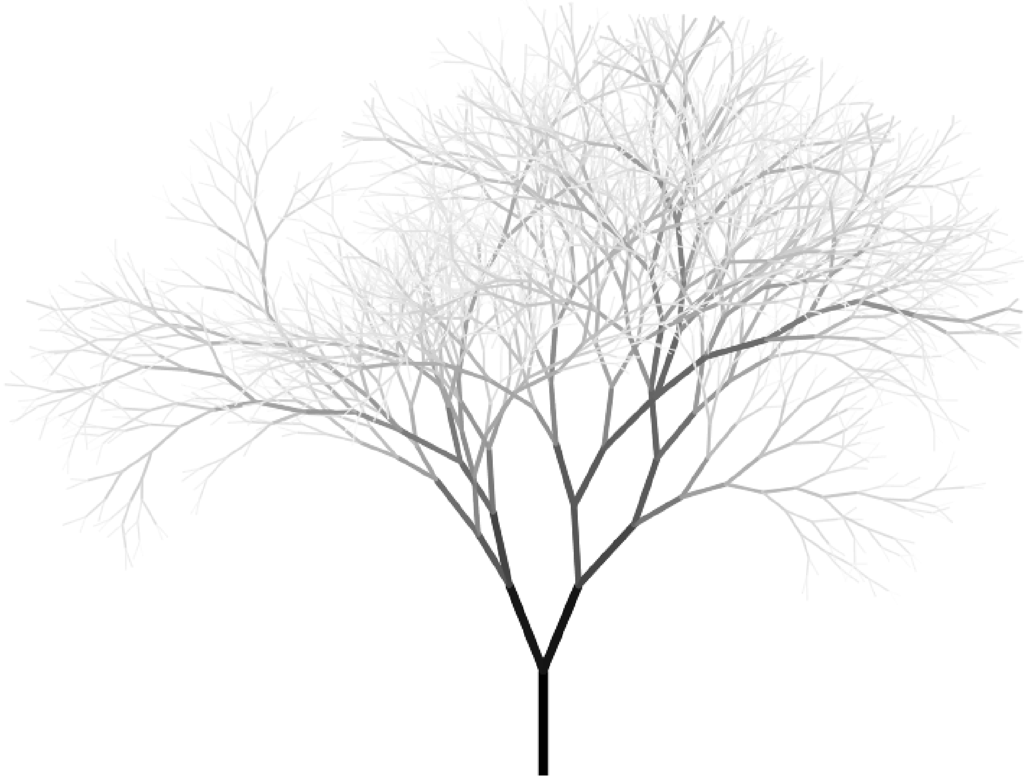
2. **Random Feature Selection**: For each tree in the forest, a random subset of features is considered at ***each split***.
   In other terms, in the same tree, we change the features we consider at each node

Label	Weight	Height	Number of legs	Class (biology)
Cat	5	10	4	Mammalia
Dog	10	15	4	Mammalia
Snake	2	3	0	Reptilia
Cat	6	12	4	Mammalia
Dog	12	18	4	Mammalia
Snake	1	2	0	Reptilia
Cat	4	8	4	Mammalia
Dog	8	14	4	Mammalia
Snake	3	5	0	Reptilia
Cat	7	14	4	Mammalia

Random Forest

How it works - example

3. **Decision Tree Construction**: A decision tree is constructed for **each bootstrap sample**.
   The tree is grown until a stopping criterion is met, such as a maximum depth or a minimum number of samples in a leaf.

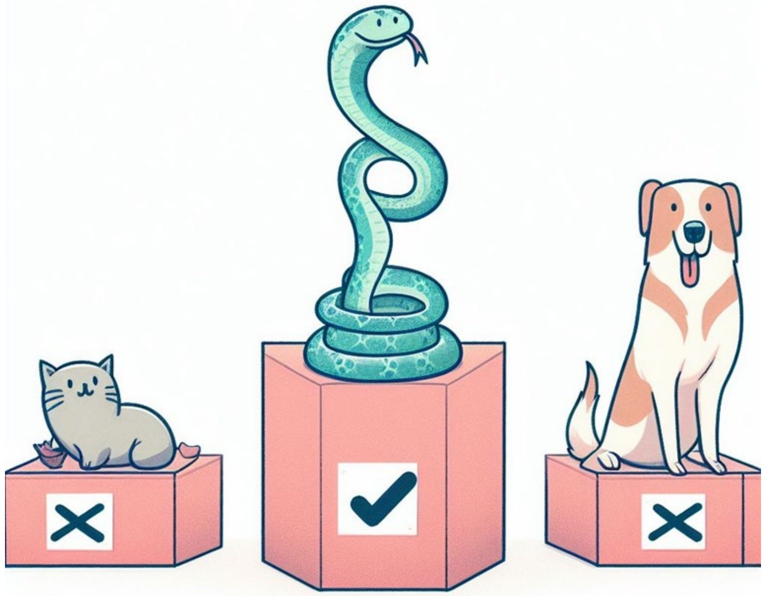


Random Forest

How to use them

- We have a new (unknown) item

Label	Weight	Height	Number of legs	Class (biology)
??	5	10	0	Reptilia



4. **Voting (for Classification) or Averaging (for Regression)**:
   - We run it through all the Decision tree we built, then…
   - For *classification* tasks, the outputs of individual trees (class predictions) are combined through a majority vote.
   - For *regression* tasks, the outputs are averaged.

Random Forest

Advantages and drawbacks

- Advantages
  - **Reduced overfitting**: Random Forests are less prone to overfitting compared to individual decision trees. The ensemble nature, with the combination of multiple trees, helps reduce variance and provides a more generalized model.
  - **High Accuracy**: Random Forests often achieve higher accuracy than individual decision trees. The combination of diverse trees can capture complex patterns in the data.
  - **Robustness to Outliers**: The averaging or voting mechanism in Random Forests can mitigate the impact of outliers present in the dataset.
  - **Feature Importance**: Random Forests can provide a measure of feature importance. This can be valuable for understanding which features contribute the most to the model's predictions.
  - **Handles Missing Values**: Random Forests can handle missing values in the dataset without the need for imputation.
  - **Parallel Training**: Training individual trees in a Random Forest can be done in parallel, leading to faster training times compared to sequentially building a single decision tree.

Random Forest

Advantages and drawbacks

- Drawbacks of Random Forest:
  - **Complexity**: Random Forests are more complex than individual decision trees, which may make them harder to interpret, especially when dealing with a large number of trees.
  - **Computational resources**: Random Forests can require more computational resources, especially as the number of trees and the size of the dataset increase.
  - **Memory usage**: The ensemble nature of Random Forests can lead to higher memory usage, as it needs to store multiple trees.
  - **Loss of interpretability**: While Random Forests provide feature importance, the interpretability of individual trees is lost, making it challenging to explain specific predictions.
  - **Not suitable for *very small* datasets**: Random Forests may not perform well on very small datasets due to the ensemble approach, which requires enough data to capture patterns effectively.

Random Forest

My personal opinion… (a.k.a., the first algo that I try)

- Random Forest is one of the best ML algorithms
  - It can do everything: Classification, Regression…
  - … it can natively handle all types of data (numerical, categorical), with no need for scaling/standardization…
  - … for a relatively low computation cost and easy parallelization

- It offers “Interpretability” (in terms of feature importance ranking)

- There are “Online” versions of it (can update the existing model with new samples without retraining on the whole dataset)

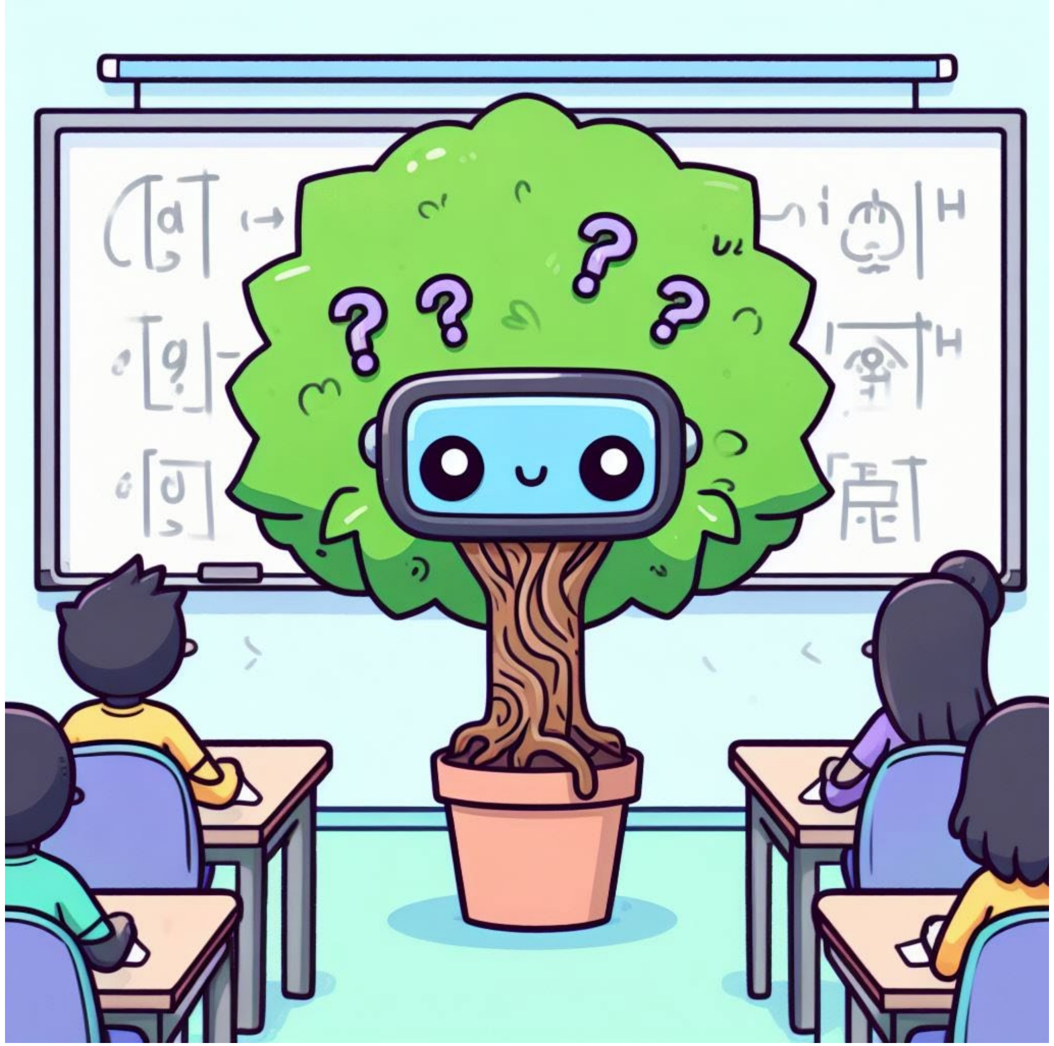
- **IMHO**: RF is the first thing to throw at a new problem especially if the dataset is “small”

Your competences after this course

You should be able to…

- … **explain** the difference between “Lazy” learning and “Eager” learning
- … **implement** a simple Decision Tree using Gini impurity and Information gain
- … **know** the main steps to build a Random Forest algorithm
- … **explain** advantages and drawbacks of Random Forest and Decision Tree approaches

Any question?



Prompt: “cute tree in a class asking machine learning questions”

A couple of questions for you!

- What is “explainable AI”?
- Which are the hyperparameters of a random forest model?
- How to use entropy instead of Gini impurity to estimate uncertainty in a decision tree?

Other resources

- Decision and Classification Trees, Clearly Explained!!!
  https://www.youtube.com/watch?v=_L39rN6gz7Y
- StatQuest: Random Forests Part 1 - Building, Using and Evaluating
  https://www.youtube.com/watch?v=J4Wdy0Wc_xQ